

**A WEB-BASED CAPSTONE AND THESIS PROJECT PROGRESS TRACKING AND
EVALUATION SYSTEM FOR STI COLLEGE ALABANG**

A Capstone Project Proposal
Presented to the Faculty of the
Information and Capstone Communications Technology Program
STI College of Alabang

In Partial Fulfillment of
the Requirements for the Degree
Bachelor of Science in Information Technology

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May 15, 2026

ENDORSEMENT FORM FOR ORAL DEFENSE

TITLE OF RESEARCH: A Web-Based Capstone and Thesis Project Progress
Tracking and Evaluation System for STI College Alabang

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In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense.

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May 15, 2026

APPROVAL SHEET

This capstone project proposal titled: **WEB-BASED CAPSTONE AND THESIS PROJECT PROGRESS TRACKING AND EVALUATION SYSTEM FOR STI College Alabang**, prepared and submitted by Belardo, Divine, Lucero, Princess, Morte, Kerby Bryan, Oba, Isha Mae, and Yanag, Janryl Carl, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

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ABSTRACT

Title of Research: **A Web-Based Capstone and Thesis Project Progress Tracking and Evaluation System for STI College Alabang**

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The main goal of this study is to create a Web-Based System for Tracking and Evaluating Capstone and Thesis Projects for STI College Alabang. This system will help make it easier to manage, monitor, and keep records of student research projects. It will solve common problems like late feedback from advisers, files being scattered across different platforms, difficulty in tracking project progress, and no single tool to monitor everything.

The system allows students, advisers, and coordinators to access it with secure accounts. Students can upload weekly journals, progress reports, and thesis documents, and the system keeps track of all revisions and submission dates. Advisers can review submissions, give feedback, approve work, or ask for changes while updating the project's milestones like Title Proposal, Chapter Approvals, and Final Defense. Coordinators can organize groups, assign advisers, set deadlines, and monitor overall progress.

The system also has automatic reminders for deadlines, notifications, and a full activity log showing uploads, approvals, feedback, and status updates. By putting all these features in one platform, the system makes communication easier, keeps everyone informed, and ensures records are accurate and organized.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF NOTATIONS

Notation	Meaning
NEXT JS	React framework for building web applications
NODE JS	JavaScript runtime environment
IDE	Integrated Development Environment
PDF	Portable Document Format
DOCX	Microsoft Word document file format
M365	Microsoft 365
VS Code	Visual Studio Code
API	Application Programming Interface
SQL	Structured Query Language
POSTGRESQL	Open-source relational database management system
SUPABASE	Backend-as-a-Service platform built on PostgreSQL
VERCEL	Cloud platform for deploying web applications
GITHUB	Web-based platform for version control and collaboration

INTRODUCTION

Project Context

Modern technology has significantly transformed the way educational institutions manage academic processes, communication, and performance monitoring, as the advancement of web-based systems, cloud computing, and real-time data management enables schools and universities to shift from manual and paper-based operations to automated digital platforms that improve efficiency, accuracy, and accessibility.

In higher education institutions such as STI College Alabang, capstone and thesis projects play a crucial role in assessing students' academic competencies, research skills, and professional readiness, yet managing project progress, monitoring milestones, tracking revisions, and evaluating submissions becomes challenging when done manually or through fragmented communication tools, often resulting in delayed feedback, lost document versions, unclear deadlines, and limited monitoring.

To address these issues, a Web-Based Capstone and Thesis Project Progress Tracking and Evaluation System is proposed to centralize project management processes into a single digital platform where students can upload documents with version control, advisers can provide structured evaluations and feedback, and coordinators can monitor group progress and milestone compliance efficiently.

The system also incorporates automated notifications, deadline monitoring, and activity logging to ensure timely submissions, improve communication, and maintain accurate documentation of academic progress while promoting transparency and accountability among students, advisers, and coordinators involved in the project.

Overall, the implementation of this system enhances efficiency, supports real-time monitoring, strengthens collaboration, and ensures that academic projects are managed in a structured, organized, and reliable manner that meets institutional standards.

According to UNESCO (2023), digital transformation in education strengthens institutional resilience, improves learning management systems, and supports more effective monitoring and evaluation processes, highlighting the importance of adopting digital solutions in academic project supervision to ensure quality education and efficient administrative operations.

In summary, the Web-Based Capstone and Thesis Project Progress Tracking and Evaluation System provides a comprehensive solution to the limitations of manual project management by centralizing document submission, feedback, and milestone tracking, thereby improving efficiency, accountability, and collaboration while ensuring that deadlines are met and academic standards are consistently maintained.

Statement of the Problem

General Problem

Managing capstone and thesis projects at STI College Alabang has become increasingly challenging for students, advisers, and coordinators due to the reliance on manual processes and fragmented communication methods. Traditional approaches, such as email exchanges, printed documents, and separate tracking tools, often result in delayed submissions, lost files, unclear deadlines, and limited visibility into project milestones. These problems create inefficiencies in monitoring and evaluation, making it difficult for advisers to provide timely feedback and for students to stay on track. Therefore, there is a clear need for a centralized system to improve organization, communication, and overall project management.

Specific Problem

- **How to design an intuitive landing page that clearly presents the system's purpose, features, and information?**

Users may have difficulty understanding the system's purpose and features immediately upon access if there is no clear and organized entry point.

- **How to develop a module enabling login for users through their Microsoft 365 credentials?**

Students may have difficulty organizing and tracking their workload if there is no centralized interface that displays their assigned projects, tasks, and ongoing activities in one place. The absence of a structured dashboard may lead to poor task monitoring and lack of progress awareness.

- **How to develop a dashboard module that allows users to efficiently view, manage, and monitor projects and tasks?**

Students may have difficulty organizing and tracking their workload without a centralized interface that displays assigned projects, tasks, and ongoing activities in one place, leading to poor task monitoring of their progress. Similarly, advisers may struggle to monitor student performance and project development without a unified system that presents project statuses, student activities, and task updates, resulting in delayed feedback and inefficient supervision.

- **How to develop a board and task management module that allows users to organize, track, and manage tasks effectively?**

Without a centralized and visual system, students may struggle to track task progress and manage details, while advisers may find it difficult to assign, monitor, and evaluate tasks, leading to disorganization, missed deadlines, and inefficient collaboration.

- **How to develop a sprint and backlog module that supports planning, organizing, and prioritizing tasks within a defined timeframe?**

Advisers may struggle to properly monitor task progress, assign responsibilities, and guide students if there is no centralized board system that visually organizes all project tasks and their statuses, resulting in inefficient supervision and delayed project completion.

- **How to develop a roadmap module that allows users to visualize project timelines, milestones, and overall progress?**

Students may have difficulty organizing and tracking individual task components such as attachments, comments, subtasks, deadlines, and assigned members if there is no dedicated system for managing task details. This may result in poor task coordination, missed deadlines, and inefficient collaboration.

- **How to develop a roadmap module that allows users to visualize project timelines, milestones, and overall progress?**

Advisers may struggle to effectively oversee task progress, provide guidance, and evaluate student output if there is no centralized system that allows them to manage task details such as descriptions, attachments, comments, subtasks, deadlines, and member assignments.

- **How to develop archive and recycle bin modules that securely store and recover project data?**

Users may have difficulty planning and tracking project progress in an organized time-based structure if there is no system for grouping tasks into sprints with defined start and end dates, goals, and scope. This may lead to unstructured workflow, delayed task completion, and poor project planning.

- **How to develop a groups module that manages members, roles, and adviser assignments effectively?**

Advisers may struggle to effectively organize project timelines and monitor grouped task progress if there is no centralized sprint system that allows them to define sprint details, assign scope items, and manage sprint objectives, resulting in less efficient project supervision and planning.

- **How to develop notification and shortcut modules that improve accessibility and keep users informed?**

Students may have difficulty understanding the overall flow of the project and tracking deadlines if there is no visual timeline that shows tasks and milestones across different phases. This may lead to poor planning, missed deadlines, and lack of awareness of project progress.

- **How to develop a profile module that allows users to manage personal information and monitor their activity?**

Advisers may struggle to track overall project progress and individual member contributions if there is no centralized timeline view that presents tasks, milestones, and project phases in a structured format, resulting in less effective monitoring and planning.

Purpose and Description

This research aims to develop a Web-Based Capstone and Thesis Project Progress Tracking and Evaluation System designed for students, advisers, and coordinators at STI College Alabang. The system is intended to improve the management, monitoring, and evaluation of capstone and thesis projects by providing a centralized digital platform. Through this system, students can upload project documents, track their progress, and receive feedback from their advisers. Advisers and coordinators will also be able to monitor project milestones, review submissions, and manage academic records efficiently. By integrating these features, the system aims to enhance communication, organization, and transparency in the management of academic research projects.

Objectives of the Project

This study aims to design and develop a Web-Based Capstone and Thesis Project Progress Tracking and Evaluation System that will support the effective supervision and monitoring of research projects at STI College Alabang. The primary objective of this project is to create a system that will help students manage their thesis or capstone requirements while enabling advisers and coordinators to evaluate project progress more efficiently. The platform will provide tools for document submission, milestone tracking, feedback management, and progress monitoring, allowing all users to collaborate effectively and ensure that academic research projects are completed in an organized and timely manner.

General Objective

The general objective of this study is to develop a Web-Based Thesis and Capstone Project Progress Tracking and Evaluation System designed to improve the management and

monitoring of research projects at STI College Alabang. The system aims to provide an efficient and organized way for students to submit thesis and capstone documents, track student progress, and receive timely feedback from advisers. In addition, the platform will allow advisers and coordinators to monitor project milestones, review submissions, and manage document revisions in a centralized system. This platform ensures that thesis and capstone projects are supervised in a structured, transparent, and efficient manner, improving collaboration and the overall research experience of students.

Specific Objectives

- **To design an intuitive landing page that clearly presents the system's purpose, features, and information.**

The proponents of the study will design a main landing page that serves as the system's entry point, providing users with a clear overview of the platform and guiding them to its main features.

- **To develop a login module that enables users to securely access the system using Microsoft 365 credentials.**

The proponents will create a features page that showcases the system's key functions in an organized manner, allowing users to easily understand tools such as progress tracking, task management, and collaboration features.

- **To develop a dashboard module that allows users to efficiently view, manage, and monitor projects and tasks.**

The proponents will design an about page that provides a clear overview of the system's purpose in improving project management and collaboration between students and advisers, presenting its role in supporting an organized workflow.

- **To develop a board and task management module that allows users to organize, track, and manage tasks effectively.**

The proponents of the study will develop a login module that allows both students and advisers to securely access the system using Microsoft 365 credentials (email and password), ensuring proper authentication, enhanced security, and controlled access for different user roles within the platform.

- **To develop a sprint and backlog module that supports planning, organizing, and prioritizing tasks within a defined timeframe.**

The proponents of the study will create a dashboard module that serves as the main interface for students, allowing them to view assigned projects, track tasks, and monitor ongoing activities. The module will help students stay organized and efficiently manage their workload. Project creation will not be available for students, as projects will be created and assigned by the adviser.

- **To develop a roadmap module that allows users to visualize project timelines, milestones, and overall progress.**

The proponents will develop a dashboard module that serves as the main interface for advisers, enabling them to view all projects, monitor student activities, and track task progress across groups. The module will also allow advisers to create and assign projects to students, ensuring efficient workflow management and supervision.

- **To develop a weekly journal module that automatically tracks and records user activities and progress.**

The proponents of the study will create a board module that enables students to manage their project tasks using a kanban-style interface. Tasks will be organized into columns such as To-do, In Progress, Revision, and Completed. Each task card will display

important details such as task title, deadline, comments, and assigned members. The module will also support task creation, but only the adviser and adviser-authorized students will be allowed to create tasks. This module will help students track progress, collaborate with group members, and stay organized.

- **To develop archive and recycle bin modules that securely store and recover project data.**

The proponents will develop a board module that enables advisers to manage project tasks using a kanban system. Advisers can create, assign, and control tasks, as well as organize them into status-based columns. Each task card will include details such as task title, code, deadline, comments, and assigned members. The module also allows advisers to control which students are authorized to create tasks, helping ensure proper supervision, progress tracking, and efficient project management.

- **To develop a groups module that manages members, roles, and adviser assignments effectively.**

The proponents of the study will create a task management module that enables students to view and manage detailed task information, including attachments, comments, and subtasks. The module allows students to break tasks into smaller parts, update task status, assign members, and monitor deadlines. This will help improve task organization, collaboration, and completion efficiency.

- **To develop notification and shortcut modules that improve accessibility and keep users informed.**

The proponents will develop a task management module that enables advisers to view and manage detailed task information within a project. Advisers can add or update task descriptions, view attachments, add comments, evaluate student work, assign members, set

deadlines, and monitor progress. This module ensures proper guidance, assessment, and efficient completion of student tasks.

- **To develop a profile module that allows users to manage personal information and monitor their activity.**

The proponents of the study will create a sprint module that enables users to manage sprints by defining the sprint name, duration, start date, and end date. The module will group tasks from the board that are ongoing or need to be completed within a specific timeframe. Users can select scope items and provide sprint goals and descriptions, helping improve planning, tracking, and overall project progress.

Significance of the Project

The implementation of the Web-Based Thesis and Capstone Project Progress Tracking and Evaluation System at STI College Alabang aims to improve the management and monitoring of research projects. The system provides a centralized platform that enhances organization, communication, and transparency among coordinators, faculty members, and students involved in thesis and capstone development.

Coordinator

The coordinator will benefit from an organized platform that allows them to monitor the overall progress of thesis and capstone projects. The system enables the coordinator to track student progress, manage project milestones, and ensure that submission deadlines are followed. In addition, the coordinator has administrative control to grant system access to faculty members, allowing them to review student submissions and provide feedback. This helps maintain proper coordination, efficient supervision, and organized documentation of all thesis and capstone projects.

Adviser

Advisers will benefit from a system that allows them to review student submissions, provide structured feedback, and evaluate thesis and capstone project progress more effectively. Through the platform, advisers can monitor revisions, check document versions, and communicate with students in an organized manner. This helps advisers guide students properly throughout the research process and ensure that academic standards are maintained.

Students

Students will benefit from a centralized system where they can upload thesis and capstone documents, track their progress, and receive feedback from their faculty advisers. The platform helps students stay informed about deadlines, revisions, and project milestones. By using this system, students can manage their research work more efficiently while maintaining better communication with their advisers and coordinators.

Future Developer/Researcher

For future researchers and developers, this system may serve as a reference for further studies and improvements related to academic project management systems. Future enhancements may include integrating advanced analytics, automated progress reports, and additional collaboration tools to further improve the monitoring and evaluation of thesis and capstone projects.

Scope and Limitations

Scope

This study's scope is to provide a fully functional web-based capstone and thesis project tracking system for STI College Alabang. The system aims to enhance the academic management process by allowing students and faculty members to monitor project progress, manage tasks, track milestones, submit revisions, and communicate efficiently. It focuses on providing a user-friendly interface, secure login, organized file management, and real-time updates to ensure smooth operation and effective supervision of academic projects.

Landing Page

- **Main Landing Page** - Serves as the system's main entry point, giving users an immediate understanding of its core purpose and the features the platform provides.

- **Features Page** - The Features Page showcases the key functionalities integrated into the system. It is designed to help users quickly explore the platform's tools for progress tracking, collaboration, and task management.
- **About Page** - The About Page provides an overview of the system and its purpose in improving project management and collaboration for students and advisers. It introduces the platform as an organized workspace that helps users track tasks, monitor progress, and manage projects more efficiently through a structured workflow.

Student Side

- **Login Module** - In this module, the user is required to enter their login credentials, including email and password, to securely access their account through Microsoft 365 login authentication.
- **Dashboard Module** - This module serves as the main interface where students can view and manage their assigned projects and tasks. It provides an overview of recent projects, worked-on and assigned tasks, and ongoing activities. The dashboard helps students stay organized, track their progress, and efficiently manage their workload within the system. Project creation is not available for students, as projects are created by the adviser and assigned to them.
- **Board Module** - This module allows students to view and manage their project tasks using a kanban board. Tasks are organized into columns based on their status, such as To-do, In Progress, Revision, and Completed. Each task card displays important details, including

the task title, and deadline, along with options for comments, and assigned members. This module also supports task creation within the board interface; however, only the adviser and students authorized by the adviser are allowed to create tasks. It helps students track their task progress, collaborate with group members, and stay organized throughout the project.

- **Task Management Module** - This module allows students to view and manage detailed information for each task within a project. It includes the task attachments, comments, and subtasks, enabling users to break down work into smaller, manageable parts. Students can update task status, assign members, and monitor deadlines. This module helps track progress, improve collaboration, and ensure tasks are completed efficiently.
- **Sprint Module** - This module allows users to manage sprints by defining the sprint name, duration, start date, and end date. It contains tasks that need to be completed or are still in progress from the board, helping organize work within a specific timeframe. Users can select scope items and provide a description of the sprint's goals and focus. This module improves planning, tracking, and overall project progress. Only the adviser can create sprints, but students may be allowed if authorized by the adviser.
- **Roadmap Module** - This module provides a timeline view of the project, presenting tasks and milestones in a chart format. It allows students to visualize the overall schedule, track progress over time, and understand how tasks are distributed across different phases. The roadmap helps students plan ahead, monitor deadlines, and ensure that project goals are completed within the expected timeframe. It also allows students to view their own progress and contributions within the project.

- **Backlog Module** - This module contains a list of all project tasks, including those from the board. It provides a centralized view where users can review, organize, and prioritize tasks before they are included in a sprint. The backlog helps ensure that all pending work is tracked and properly planned for future development.
- **Weekly Journal Module** - This module automatically generates a formatted record of tasks completed and activities performed within a week. It collects data from the system, allowing students to view their weekly progress without manually creating a journal. This helps students stay organized and easily track their work and accomplishments.
- **Archive Module** - This module stores completed or inactive project items, such as finished tasks and closed sprints, for future reference. It allows users to keep a record of past work without affecting active tasks on the board. This helps maintain an organized workspace while preserving important project history that can be reviewed when needed.
- **Recycle Bin Module** - This module allows students to recover or restore project related items they have previously deleted, such as tasks and other project data. It serves as a safety feature to prevent accidental loss by enabling users to retrieve important items that were mistakenly removed. Deleted items are stored temporarily and will be permanently removed from the system after a set period, encouraging proper data management while maintaining system efficiency.
- **Groups Module** - This module manages group information, including group members and adviser assignments. It allows users to view the list of members in each group and identify

the assigned adviser. This module helps organize team structure and ensure proper coordination between students and advisers.

- **Shortcut's URL Module** - This module provides students with quick access to important external platforms, such as ELMS and One STI. These shortcuts are displayed as clickable links, allowing students to easily navigate academic resources outside the system.
- **Notification Module** - This module provides users with real-time updates and alerts related to their projects and tasks. It notifies students about important activities such as task assignments, updates, deadlines, and system announcements. This helps users stay informed, respond promptly, and keep track of all relevant actions within the system.
- **Profile Module** - This module allows students to view their personal information, including their name and school email, along with an overview of their activities across multiple projects. It displays basic information such as assigned and completed tasks. The module may also include a simple chart to track activity over time. Additionally, it provides a feedback feature where students can rate their experience and submit or edit feedback about the system.

Adviser Side

- **Login Module** - In this module, the user is required to enter their login credentials, including email and password, to securely access their account through Microsoft 365 login authentication.

- **Dashboard Module** - This module serves as the main interface where advisers can view and manage all projects and assigned tasks. It provides an overview of recent projects, student activities, and ongoing tasks across groups. The dashboard helps advisers monitor progress, track student performance, and efficiently manage project workflows. Advisers also have the ability to create and assign projects to students.
- **Board Module** - This module allows advisers to view and manage project tasks using a kanban board. Tasks are organized into columns based on their status, such as To-do, In Progress, Revision, and Completed. Each task card displays important details, including the task title, code, deadline, comments, and assigned members. Advisers can create, assign, and manage tasks within the board, as well as controlling which students are authorized to create tasks. This module helps advisers monitor task progress, guide students, and ensure project completion.
- **Sprint Module** - This module allows advisers to create and manage sprints by defining the sprint name, duration, start date, and end date. It includes tasks from the board that need to be completed or are still in progress, helping organize work within a specific timeframe. Advisers can select scope items and provide a description of the sprint's goals and focus. This module supports planning, tracking, and overall project progress, with full control over sprint creation and management.
- **Task Management Module** - This module allows advisers to view and manage detailed information for each task within a project. It includes the task description, attachments, comments, and subtasks, enabling work to be broken down into smaller parts. Advisers can add or update descriptions, view attachments, add comments, evaluate student work, assign

members, set deadlines, and monitor progress. This module helps ensure tasks are properly guided, assessed, and completed efficiently.

- **Sprint Module** - This module allows advisers to create and manage sprints by defining the sprint name, duration, start date, and end date. It includes tasks from the board that need to be completed or are still in progress, helping organize work within a specific timeframe. Advisers can select scope items and provide a description of the sprint's goals and focus. This module supports planning, tracking, and overall project progress, with full control over sprint creation and management.
- **Roadmap Module** - This module provides a timeline view of the project, presenting tasks and milestones in a chart format. It allows advisers to visualize the overall schedule, track progress over time, and understand how tasks are distributed across different phases. The roadmap helps advisers monitor deadlines and ensure that project goals are completed within the expected timeframe. It also enables advisers to monitor the progress and contributions of individual members.
- **Backlog Module** - This module contains a list of all project tasks, including those from the board and sprints. It provides a centralized view where advisers can review, organize, and prioritize tasks before they are included in a sprint. The backlog helps advisers track pending work, manage task distribution, and ensure proper planning for ongoing and future development.
- **Weekly Journal Module** - This module allows advisers to view and monitor students' weekly journals, which are automatically generating records of tasks completed, and activities performed within a week. It collects data from the system, enabling advisers to track each student's progress without requiring manual journal creation. This helps

advisers assess performance, monitor contributions, and ensure consistent progress throughout the project.

- **Archive Module** - This module stores completed or inactive project items, such as finished tasks, closed sprints, and archived projects, for future reference. It allows advisers to keep a record of past work without affecting active tasks on the board. This helps maintain an organized workspace while preserving important project history that can be reviewed when needed. Advisers can also archive projects through this module.
- **Recycle Bin Module** - This module allows advisers to recover or restore project related items they have previously deleted, such as tasks, projects, and other data. It serves as a safety feature to prevent accidental loss by enabling retrieval of important items. Deleted items are stored temporarily and will be permanently removed from the system after a set period, encouraging proper data management while maintaining system efficiency.
- **Groups Module** - This module manages group information, including group members and adviser assignments. It allows advisers to view and manage the list of members in each group, assign or confirm advisers, and oversee group composition. Advisers also have control to authorize students to create tasks and manage group roles. This module helps organize team structure and ensure proper coordination within the group.
- **Shortcut's URL Module** - This module provides advisers with quick access to important external platforms, such as ELMS and One STI. These shortcuts are displayed as clickable links, allowing students to easily navigate academic resources outside the system.

- **Notification Module** - This module provides advisers with real-time updates and alerts related to projects and tasks. It notifies advisers about important activities such as task progress, student updates, deadlines, and system announcements. This helps advisers stay informed, monitor ongoing work, and respond promptly to ensure smooth project management.
- **Profile Module** - This module allows advisers to view and manage their personal information, including their name and school email. It may also display an overview of their activity within the system. Additionally, it provides a feedback feature where advisers can rate their experience and submit or edit feedback about the system.

Limitation of the Project

- TrackSphere is accessible only to STI College Alabang students and faculty. Only users with valid STI College Alabang credentials can log in, as the system supports Microsoft 365 authentication only.
- Students must register in the system at least once before they can be assigned to a project. The system does not support automatic user creation, so a student must first have an existing account record before being added by an adviser.
- The system is limited to Capstone and Thesis project management and does not cover other academic processes like course enrollment or grading.
- The file upload feature supports PDF, DOCX, and TXT files only, with a maximum file size of up to 200 MB per upload.
- Deleted items in the Recycling Bin are retained for up to thirty days, giving users time to review and restore them if needed. After this period, the system automatically removes the items to maintain storage efficiency and ensure smooth performance.

REVIEW OF RELATED LITERATURE

Review of Related Literature

The Review of Related Literature examines previous studies and existing systems that form the basis for developing the Web-Based Thesis and Capstone Project Progress Tracking and Evaluation System at STI College Alabang. It explores project management, academic monitoring, and collaboration platforms to identify best practices and features that can improve supervision, communication, and progress tracking for thesis and capstone projects.

Local Literature

Web-Based Document Tracking System for Catanduanes State University (2025)

(Acedo, Bola, and Morales, 2025) designed a web-based document tracking system for Catanduanes State University to enhance administrative efficiency and transparency in handling academic documents. Manual processes were found to cause misplaced files, delays, and errors. The system uses a centralized database and QR-code technology to uniquely identify documents for real-time status updates. Developed with HTML, CSS, JavaScript, and PHP, it allowed users to track documents instantly, minimized human error, and improved administrative workflow. Evaluation using ISO/IEC 25010 criteria confirmed high functional suitability, usability, and performance efficiency. This study informs the proposed capstone progress tracking system by demonstrating the effectiveness of web-based monitoring, centralized storage, and real-time status updates for managing academic records in Philippine educational institutions.

Design and Develop a Thesis and Capstone Manuscript Archiving Management System (2024)

Perante, Barnaba, Cañete, and Niegas (2024) created a web-based manuscript archiving management system at Eastern Visayas State University–Ormoc Campus. The system addressed challenges with manual document storage, limited accessibility, and difficulty retrieving past research. Using standard web technologies, the platform enables online submission, categorized storage, searchable access, and secure long-term preservation of thesis and capstone projects. Evaluation showed improved organization, faster document retrieval, and enhanced security compared to traditional archives. This study is relevant to the proposed capstone progress tracking system because it demonstrates the benefits of centralizing academic project data digitally. Its design supports integrating progress monitoring, evaluation, and archival functions within a single platform, improving efficiency and accessibility for both students and faculty.

STEM Web-Based System for Students' Profiling and Archiving (2023)

The STEM Web-Based System (2023) was developed to profile students and archive their research outputs digitally in a Philippine institution. The platform aimed to organize student records, store research documents, and provide efficient search and retrieval functions. Built using PHP and MySQL, the system improved record management for both students and faculty. Evaluation focused on functionality, reliability, usability, and accessibility, showing that the web-based platform significantly improved administrative efficiency compared to manual systems. Faculty could monitor students' progress more effectively, while students benefited from convenient access to their archived outputs. This study supports the development of a capstone and thesis progress tracking system by demonstrating how centralized digital platforms enhance transparency, structured record keeping, and institutional efficiency in managing student academic outputs.

Web-Based School Management System (AMA University, 2023)

AMA University (2023) developed a web-based school management system to digitize student records, faculty data, and administrative processes in a Philippine high school. The system consolidated academic and administrative information into a centralized online dashboard, improving accessibility, data accuracy, and efficiency. Manual record-keeping previously caused delays, redundancy, and errors in retrieving student information. The platform allows dynamic updates, real-time access, and secure storage of data. User evaluations indicated high satisfaction with usability and performance. This study is relevant to the proposed Web-Based Capstone and Thesis Project Progress Tracking and Evaluation System because it demonstrates the practical benefits of centralized web-based management for educational institutions, including improved monitoring of student progress and streamlined record management.

Orozco, M. M. (2023). Web-based thesis/capstone project defense evaluation system of the CCS Biñan [Undergraduate thesis, University of Perpetual Help System Laguna]. University of Perpetual Help System Laguna.

Web-Based Thesis/ Capstone Project Defense Evaluation System of the CCS Biñan (2023)

Orozco (2023) introduced a web-based system for evaluating thesis and capstone defenses at the University of Perpetual Help System Laguna, which allowed advisers and panelists to submit grades online, automatically calculate results, and securely store records in a centralized database. By replacing paper-based forms and manual computations, the system reduced errors, delays, and administrative workload. Assessment using ISO 9126 standards showed that it was functional, reliable, and easy to use. This demonstrates that web-based platforms can improve efficiency, accuracy, and transparency in academic evaluation processes, providing strong support for the

proposed Web-Based Capstone and Thesis Project Progress Tracking and Evaluation System. implementing a similar system can help streamline project monitoring, enable timely feedback, and enhance overall project management for students and advisers.

Foreign Literature

Development of Digital-Based Academic Services in Higher Education (2025)

As emphasized by Waruwu et al. (2025), digital transformation in academic services is essential for ensuring responsive, accessible, and efficient processes in higher education institutions. Their study developed a digital-based thesis defense academic service system using the ADDIE model and mixed-method approaches. Through interviews, observations, and surveys involving students and lecturers, the researchers found that the system significantly improved service quality, administrative efficiency, and user satisfaction. Features such as online registration, document submission, and transparent data management simplified academic workflows and enhanced accessibility. The success of the system led the authors to recommend its adoption as a flagship initiative for modernizing academic services in universities.

Implementation of Docker Platform for Thesis System Integration (2024)

ITScience et al. (2024) investigated the integration of thesis-related academic systems using modern containerization technology. In their study at Sumbawa University of Technology, the researchers identified inefficiencies in thesis registration, proposal scheduling, and seminar processes, which were still conducted manually despite the existence of two separate but unintegrated systems: one for thesis title checking and another for project scheduling. Using Docker and the Software Development Life Cycle (SDLC) approach, the researchers successfully merged these systems into a unified platform. This integration improved administrative efficiency,

user access, and system usability. They also recommended further enhancements in data security, system maintenance, and user-based improvements to ensure long-term system resilience.

Using Webpages to Document and Assess Student Capstone Project Work (2022)

Garry (2022) explains that capstone courses in Engineering Technology programs, as required by ABET-TAC standards, rely on direct evidence of student work for outcome assessment. At South Dakota State University, group project webpages have long been utilized to support collaboration and documentation throughout students' capstone projects. During a transitional period when webpage support software was unavailable, the researchers observed a decline in the quality of project reports. Rubric-based analysis confirmed that the absence of web-based documentation negatively affected assessment outcomes. This finding reinforced the value of digital platforms in organizing student output, facilitating collaboration, and supporting the evaluation of project-based learning.

Web-Based Master Student Project Evaluation and Management System (2022)

According to Juliana Wahid, Ng Kar Song, and colleagues (2022), the Project in Computing course under the Master of Science in Information Technology (MSc IT) program at the University Utara Malaysia serves as a vital platform for students to demonstrate their acquired knowledge and technical skills through project implementation. However, the current evaluation process remains largely manual, with students submitting proposals and articles to evaluators via email. This method is inefficient and poses challenges for coordinators, evaluators, and supervisors in organizing and monitoring project evaluations. To address these limitations, the researchers developed the "Evaluation Virtual Assistant" (EVA), a web-based project evaluation and management system designed to streamline proposal, article, presentation, and final report assessments. With multiple evaluators involved in different evaluation phases, EVA provides

structured workflows, recommendation tracking, and automated computation of results. Ultimately, the system enhances evaluation efficiency, accuracy, and consistency.

Evaluating a Structured Online Peer Evaluation System Among Graduate-Level Capstone Students (2022)

Wilkinson et al. (2022) highlight the growing challenges of online learning, particularly the lack of collaboration and interaction that negatively affects student performance and engagement. Peer review activities are known to support critical thinking and build academic communities, yet many students lack the skills and tools needed to participate effectively. To address this gap, the authors implemented a structured online peer evaluation system supported by an interactive digital toolkit for graduate communication capstone students at the University of North Coast Muscari. Grounded in Vygotsky's Zone of Proximal Development, Constructivist learning principles, and cognitive apprenticeship models, this system enabled meaningful peer interaction. Findings demonstrated that structured digital evaluation tools helped reduce anxiety, improved social engagement, and fostered a more responsible and focused peer learning environment.

SYNTHESIS

The reviewed local and foreign literature clearly establishes the importance of web-based systems in addressing the challenges associated with managing thesis and capstone projects. Across the studies, a consistent issue identified is the reliance on manual processes and fragmented communication platforms, which often result in delayed feedback, disorganized records, difficulty in tracking progress, and inefficient coordination among students, advisers, and coordinators. These limitations justify the need for a more centralized, automated, and integrated system.

Local literature strongly supports the effectiveness of web-based platforms in improving academic management processes. The study of Orozco (2023) demonstrates how a web-based evaluation system enhances accuracy, reduces delays, and ensures transparency through automated grading and centralized data storage. Similarly, Acedo et al. (2025) emphasize the importance of real-time document tracking and centralized databases in minimizing errors and improving workflow efficiency. Perante et al. (2024) further highlights the value of digital archiving systems in ensuring secure storage, easy retrieval, and better organization of academic documents. Additionally, the STEM Web-Based System (2023) and AMA University (2023) confirm that centralized digital platforms significantly improve accessibility, usability, and overall administrative efficiency, particularly in monitoring student progress and managing academic records.

Foreign literature complements these findings by emphasizing structured workflows, collaboration, and system integration. Wahid et al. (2022) introduced a web-based evaluation system that improves consistency and efficiency through automated processes and organized workflows. Wilkinson et al. (2022) highlights the role of digital tools in enhancing collaboration and engagement among students, while Garry (2022) demonstrates that web-based documentation contributes to better quality outputs and more effective assessment. Furthermore, Waruwu et al. (2025) stress the importance of digital transformation in improving academic services, and

ITScience et al. (2024) show that integrating multiple systems into a unified platform enhances usability and operational efficiency.

Despite these advancements, most existing systems focus only on specific functions such as evaluation, document tracking, or archiving, leaving a gap in providing a fully integrated solution. The proposed Web-Based Capstone and Thesis Project Progress Tracking and Evaluation System addresses this gap by combining essential features identified in the literature, including milestone tracking, task management, document version control, real-time monitoring, and automated notifications. By integrating these functionalities into a single platform, the system ensures improved efficiency, transparency, and collaboration. Therefore, the development of the proposed system is justified as it not only builds upon existing solutions but also enhances them by providing a more comprehensive and unified approach to academic project management.

Technical Background

Overview of Current Technologies to be Used in the System

The proposed system is developed as a web-based capstone and thesis project tracking and monitoring platform for the academic community of Sti College Alabang. It is designed to provide an efficient and organized way for students, advisers, and coordinators to manage project progress, monitor milestones, and evaluate submissions throughout the research process.

The system aims to improve communication, ensure timely feedback, and maintain accurate documentation of project activities. It allows users to upload files, track revisions, and receive updates in real time, reducing delays and miscommunication.

The platform is built using modern web technologies commonly used in web development, enabling accessibility through browsers without requiring installation. It also ensures secure data handling, role-based access, and a user-friendly interface, making the system reliable, scalable, and effective for academic project management.

Next.js

The developers will use Next.js as the primary framework for building the web-based system, enabling fast rendering and efficient routing. It supports server-side rendering and improves performance, scalability, and overall user experience.

TypeScript

The developers will use TypeScript to enhance code quality and maintainability. It provides type safety, reduces errors, and improves development efficiency by enabling better structure and debugging during system development.

Tailwind CSS

The developers will use Tailwind CSS to design the system's interface, ensuring a consistent, responsive, and visually appealing layout. It allows rapid styling using utility classes and improves overall user experience.

Shadcn/UI

The developers will use Shadcn/UI to implement pre-designed and customizable user interface components. It ensures consistency, accessibility, and faster development of modern and clean user interfaces.

Reactbits

The developers will use Reactbits to enhance component functionality and improve user interaction. It provides reusable logic and utilities that support efficient development and better performance within the system.

Node.js

The developers will use Node.js as the backend runtime environment to handle server-side operations. It manages requests, processes data, and ensures efficient communication between the frontend and the database.

PostgreSQL

The developers will use PostgreSQL as the primary database to store and manage system data. It ensures data integrity, reliability, and efficient handling of structured data such as user accounts and project records.

Supabase

The developers will use Supabase as a backend service to provide database management, authentication, and real-time capabilities. It simplifies backend development and ensures secure and scalable data operations.

Vercel

The developers will use Vercel for deploying the system, enabling fast and reliable hosting. It provides continuous deployment, scalability, and optimized performance for modern web applications.

Visual Studio Code (VS Code)

The developers will use Visual Studio Code as the integrated development environment for building and managing the system. It provides essential tools such as code editing, debugging, and extensions that support efficient development using modern web technologies. Visual Studio Code enhances productivity through features like syntax highlighting, auto-completion, and version control integration, making the development process more organized and efficient.

Calendar of Activities

Brainstorming

In this phase, the developers begin by thinking deeply about the project and exploring possible ideas for the system. The team discusses different concepts, features, and approaches that can be applied to meet the objectives. They analyze the needs of the users and identify problems that the system aims to solve. Initial plans, including target dates and task outlines, are also considered during this stage. Additionally, potential challenges and required resources are identified, and roles are tentatively assigned to ensure a clear direction for the development process.

Interview

In order to clearly identify the client's needs, the developers conducted interviews from the fourth week of February until the first week of June. This process helped them gather detailed information about the desired features, system modules, and other important specifications. It also allowed the team to clarify expectations, identify possible challenges, and confirm system requirements. Through continuous communication with the client, the developers ensured that the proposed system design is aligned with the client's needs and project objectives.

Documentation: Chapter 1

The team developed this chapter from the second week of February until last week of. It includes the study's introduction, statement of the problem, purpose and description, objectives, as well as the scope and limitations of the study. Revisions were made whenever feedback and recommendations were provided by the advisers to improve the content and clarity of the chapter.

Documentation: Chapter 2

This chapter was prepared by the team from the fourth week of March until the end of May. It contains the review of related literature, review of related studies, and system synthesis, which serve as the foundation and background for the development of the system.

Gantt Chart of Activity

MONTH														
ACTIVITY	FEBRUARY				MARCH				APRIL				MAY	
REQUIREMENTS PHASE														
Brainstorming														
Calendar of Activities / Recourses														
Interview														
Project Context														
Purpose and Description														
Scope and Limitation														
Review and Related Literature/System														
DESIGN PHASE														
Flow Charts / Wireframe														
Design Review														
IMPLEMENTATION														
Website/System Development														
TESTING PHASE														
Fixing Error / Bug Fixing														
Testing														
DEPLOYMENT PHASE														
Deployment														

Table 1: Gantt Chart

	Complete Task
	Ongoing Task

RESOURCES

Resources Management

Software

Software Requirements	
Operating System	Windows 10 or higher
Integrated Development Environment (IDE)	Visual Studio Code
Database	PostgreSQL, Supabase
Programming language	TypeScript
Browser	Google Chrome, Edge
Documentation	Word and M365 Word

Table 2. Software Requirements for Development

Hardware Requirements	
Processor	Intel Core i5 / Ryzen 5
Operating System	Windows 10 or higher
RAM	8GB or higher
Storage	512 SSD
Display	1920x1080

Table 3. Hardware Requirements for Development

MINIMUM HARDWARE REQUIREMENTS (FOR DEVELOPMENT)

MINIMUM	
Processor	Intel Core i3 (8th gen)
Operating System	Windows 10 or higher
Memory	8GB

Graphics	GT 730 – 2GB
Storage	512 SSD

Table 3.1 Minimum Hardware Requirements for Development

MINIMUM HARDWARE REQUIREMENTS (FOR USERS)

MINIMUM	
Processor	Intel Core i3 (6th Gen or above)
Operating System	Windows 10 or higher
Memory	4 GB (8 GB recommended)
Graphics	GT 730 – 2GB
Storage	256 HDD (SSD recommended)

Table 3.2 Minimum Hardware Requirements for Users

METHODOLOGY, RESULTS, AND DISCUSSION

Iterative Waterfall Methodology

For the development of TrackSphere, the team adopted the Iterative Waterfall methodology. This approach follows a structured sequence of stages such as requirements gathering, design, implementation, testing, deployment, and review and improvement, while allowing feedback between phases for necessary revisions. It is suitable for projects with defined goals but may require continuous refinement. The Iterative Waterfall model was chosen for its organized structure that ensures proper documentation and a clear understanding of requirements before progressing, while its iterative nature allows the team to improve features, address issues, and enhance the system throughout development.

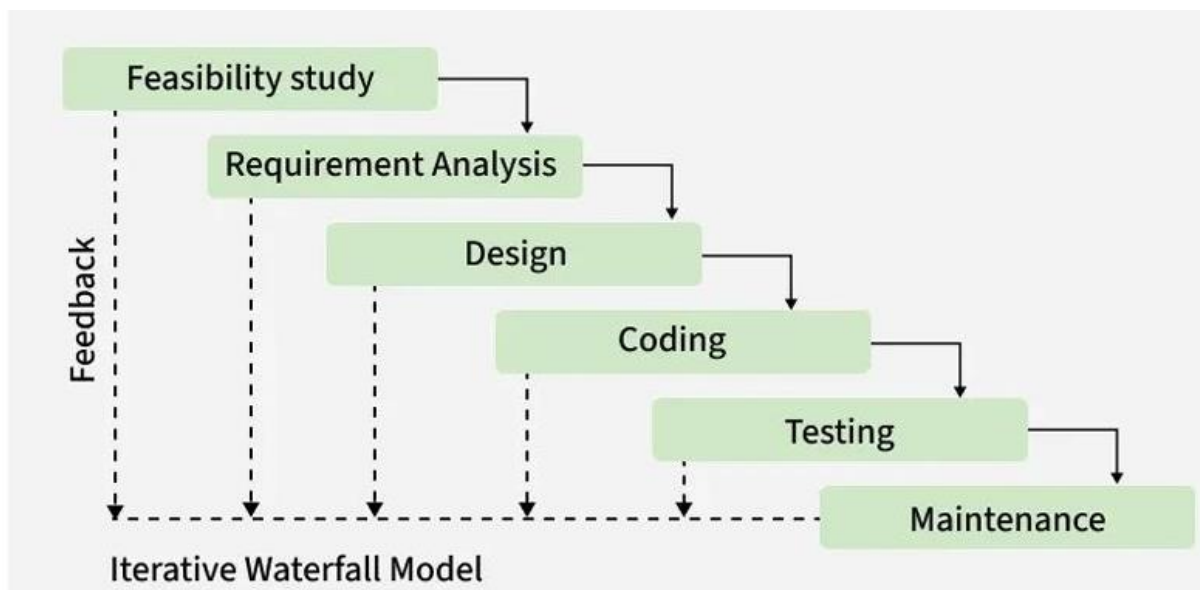


Figure 1. Iterative Waterfall Methodology Diagram

Phase 1: Requirements Gathering

The team gathered detailed requirements for TrackSphere by identifying project goals, user needs, and system functionalities. This included both functional and non-functional requirements, which were clearly documented to guide the development process.

Phase 2: Design

Based on the gathered requirements, the team created the system design, including the architecture, database structure, wireframes for the user interface, and flowcharts to illustrate system processes. The design was reviewed and revised based on feedback to ensure an effective and well-organized system structure.

Phase 3: Implementation

Developers began coding the system based on the approved design. Modules were developed step by step, including task management, board, sprint, and other system features. Any issues encountered during development were used as feedback for improving the design.

Phase 4: Testing

The system was tested to ensure it met the defined requirements. Bugs and errors were identified early, and necessary fixes were applied through feedback loops to improve system performance, usability, and reliability.

Phase 5: Deployment

Phase 6: Review and Improvement

- A. Requirements Analysis**
- B. Requirements Documentation**
- C. Design of Software, System, Product, and/or Processes.**